Curriculum map — actual

Advanced Statistics 2027-1 · Actual material reached in class and the student notebook

This is the **actual-teaching counterpart** of the planned curriculum map. Both use the same thirteen meetings and six-column lecture/source/notebook structure. The planned map records intended content and source coverage; this map records **only** what actually happens. A drafted deck, a solved notebook or an approved scope is preparation, not actual coverage. No meeting has been taught yet, so the table begins deliberately empty. After each meeting, update the corresponding row with topics substantively taught (not merely mentioned), Murphy Book 2 and Book 1 sections **actually reached**, and what students were assigned to do or attempted in the notebook. Record incomplete work, omitted material, deviations from the plan, and planned carry-forward. Distinguish an assigned exercise from observed student completion. Do not automatically copy planned topics into this table.

Lecture-by-lecture actual coverage

No.	Lecture / status	Actually taught / worked	Book 2 — reached	Book 1 — reached	Notebook assigned / attempted; omissions and follow-up
1	**Probabilistic ML and Gaussian systems** Not yet taught	—	—	—	Not yet attempted
2	**Exponential families and learning from sufficient statistics** Not yet taught	—	—	—	Not yet attempted
3	**Graphical models: factorization and conditional independence** Not yet taught	—	—	—	Not yet attempted
4	**Information theory: distributional comparison and information preservation** Not yet taught	—	—	—	Not yet attempted
5	**Generalized linear models, automatic differentiation and SGD** Not yet taught	—	—	—	Not yet attempted
6	**Latent-variable learning and expectation-maximization** Not yet taught	—	—	—	Not yet attempted
7	**Bayesian computation: grid reference and Laplace approximation** Not yet taught	—	—	—	Not yet attempted
8	**Exact inference in structured sequence models** Not yet taught	—	—	—	Not yet attempted
9	**Variational inference: posterior geometry and pathwise gradients** Not yet taught	—	—	—	Not yet attempted
10	**Monte Carlo and MCMC: invariance, mixing and estimator**	—	—	—	Not yet attempted

No.	Lecture / status	Actually taught / worked	Book 2 — reached	Book 1 — reached	Notebook assigned / attempted; omissions and follow-up
	error Not yet taught				
11	**Neural conditional models: heteroskedastic likelihood and regularization** Not yet taught	—	—	—	Not yet attempted
12	**Variational autoencoders: amortized inference and rate–distortion** Not yet taught	—	—	—	Not yet attempted
13	**Project presentations and course synthesis** Not yet taught	—	—	—	Not yet attempted